
Second Spectrum: Premier League – Player Fitness Output Formats

File Documentation Details
Version 0.3

Version History

Version	Date	Notes
v0.1	4/15/19	Initial Version
v0.2	5/29/19	Added period splits, added time as a metric, corrected field names.
v0.3	3/1/23	Consolidated match fitness files into two .csv files. Updates to data fields.

Overview

Second Spectrum will deliver two match fitness files in CSV format ~30 minutes post match. These files will be available to download in DVMS.

Output Formats

Output	Format	Message Contents	Notes
physical-splits.csv	CSV	5 minute player and team match fitness splits	Single CSV file with all game players and team aggregates for each 5 minute split. All distances are in metres.
physical-summary.csv	CSV	Player match fitness aggregates.	Single CSV file with all game player aggregates. All distances are in metres.

Physical Splits

Physical Splits files in .csv formats will be available on DVMS. Files will be delivered for all matches, post match ~30 minutes. Referees are not included in the Physical Splits file. All match players (starters and substitutes) will appear in the Physical Splits file.

The Physical Splits file is organized as follows:

1. Match Description, date and Opta game ID
2. Speed Threshold Key (km/h)
3. Home Team aggregate 5 minute Physical Splits
4. Home Team individual player 5 minute Physical Splits
5. Away Team aggregate 5 minute Physical Splits
6. Away Team individual player 5 minute Physical Splits

Example Physical Splits CSV (*imitation data*):

```
"Second Spectrum Split Data"
"ACM - TOT : 2023-2-14"
None

"Threshold key (km/h)"
"Walking : 7"
"Jogging : 15"
"LowSpeedRunning : 20"
"HighSpeedRunning : 25"
"LowSpeedSprinting : >25"

"Minute Splits",5,10,15,20,25,30,35,40,45,50, ,50,55,60,65,70,75,80,85,90,95,100
"AC Milan (120)"
"Total
Distance",6447.23,6929.51,5834.71,3617.55,4824.93,6400.69,4620.98,5702.2,6551.73,3866.2,,461
6.39,6509.62,3821.92,5873.38,5152.4,6640.39,5309.2,5021.58,5448.88,5714.18,1475.75
"Walking
Distance",1251.37,1837.45,1841.79,2128.31,1580.23,1822.41,1818.8,1992.19,1834.53,2014.95,,16
77.61,1630.79,1933.74,1764.94,1854.41,1936.08,1846.29,2127.21,2089.82,1876.98,602.12
"Jogging
Distance",3527.42,3344.9,2525.82,441.54,1913.59,2861.85,1449.84,2126.93,2657.17,1365.6,,1930
.0,3114.33,1566.18,2743.78,2062.65,3235.52,1887.94,2060.04,2197.31,2415.01,653.39
"Low Speed Running
Distance",996.83,1235.45,745.9,300.3,707.6,1110.46,705.83,834.45,1183.12,339.8,,730.68,994.5
4,442.76,297.52,799.86,1014.73,851.6,566.24,681.92,255.63,167.8
"High Speed Running
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Distance",369.3,428.79,284.95,188.79,422.29,433.42,305.87,538.58,480.2,110.25,,194.42,512.67
,152.79,328.08,342.81,332.15,333.21,216.17,327.11,412.55,42.86
"Sprinting
Distance",72.51,82.53,136.25,48.62,201.22,172.55,20.64,210.05,396.7,35.6,,53.68,357.29,26.44
,38.66,92.87,121.92,120.16,51.92,132.72,154.01,9.07
"Walking
Count",215,197,227,82,138,194,141,166,193,126,,131,201,109,194,156,220,149,164,184,187,46
"Jogging
Count",294,287,295,105,199,274,204,241,276,170,,190,286,138,274,222,298,211,211,239,259,57
"Low Speed Running
Count",137,134,102,45,90,139,95,128,131,50,,81,137,62,124,100,118,92,75,90,125,20
"High Speed Running Count",36,42,29,15,44,48,29,54,59,15,,21,51,19,35,35,36,34,23,29,47,6
"Sprinting Count",7,6,6,3,13,8,2,15,17,3,,6,14,3,4,8,6,8,6,8,12,1

"Theo Hernandez (222784)"
"Total
Distance",593.39,553.64,515.8,327.59,415.63,590.99,384.95,489.82,569.11,326.13,,381.47,535.2
9,316.32,536.7,510.48,602.55,519.17,476.88,480.63,508.46,148.51
"Walking
Distance",113.29,212.39,357.42,152.07,108.49,163.45,158.19,155.5,169.97,180.79,,89.72,143.87
,144.46,155.79,123.65,190.33,155.41,168.78,179.37,176.15,39.47
"Jogging
Distance",326.17,193.49,912.14,113.61,178.04,231.74,151.02,139.1,212.71,144.67,,191.67,282.3
8,116.62,248.68,220.39,233.97,188.78,200.32,211.46,198.6,80.52
"Low Speed Running
Distance",53.13,116.02,31.8,24.39,41.47,98.28,54.11,80.29,139.5,0.66,,81.11,60.77,28.96,86.4
8,80.64,92.68,119.81,56.89,67.97,79.12,28.52
"High Speed Running
Distance",67.92,24.53,46.85,26.43,67.0,55.83,21.63,60.06,46.93,0.0,,13.63,29.59,26.29,34.02,
85.81,22.97,41.66,31.37,25.74,48.36,0.0
"Sprinting
Distance",33.17,7.22,61.1,11.08,20.63,41.68,0.0,54.87,0.0,0.0,,5.13,68.68,0.0,11.12,0.0,61.6
1,13.5,19.53,6.09,6.1,1.0
"Walking Count",20,20,39,9,15,20,11,13,19,13,,11,20,10,16,14,19,11,13,19,18,5
"Jogging Count",23,26,21,11,19,27,17,23,34,14,,20,30,12,25,21,24,18,16,24,28,5
"Low Speed Running Count",12,21,9,6,8,12,3,14,12,0,,8,30,5,13,11,8,8,8,9,12,2
"High Speed Running Count",3,4,5,3,4,8,2,5,6,0,,2,5,2,5,3,3,3,2,4,0
"Sprinting Count",4,3,3,1,1,2,0,3,3,0,,1,3,0,1,0,1,1,2,1,3,0

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To Note:

- The numbers in parentheses immediately following the team and player names are the Opta team and player IDs

Physical Summary

Physical Summary files in .csv formats will be available on DVMS. Files will be delivered for all matches, post match ~30 minutes. Referees are not included in the Physical Summary file. All match players (starters and substitutes) will appear in the Physical Summary file.

The Physical Summary file is organized as follows:

1. Opta game ID
2. Total game time, effective playing time *EPT*, home EPT and away EPT (split by full game, first half and second half)
3. Player fitness summary - included fields:
 - a. Opta player ID, player name, total minutes played, total distance, walking distance, jogging distance, running distance, high speed running distance, sprinting distance, number of high intensity runs, top speed, average speed, distance TIP, HSR distance TIP, sprinting distance TIP, number of high intensity runs TIP, distance OTIP, HSR distance OTIP, sprint distance OTIP, number of high intensity runs OTIP, distance BOP, HSR distance BOP, sprinting distance BOP, number of high intensity runs BOP

**TIP = in possession, OTIP = out of possession, BOP = out of play, HSR = high speed run, EPT (effective playing time) = the time during a game where a ball is considered live. Excludes when the ball is out of play or dead.*

Example Physical Summary CSV (*imitation data*):

```
, "Second Spectrum Physical Summary"
, "Game ID: 2292824"

,,Total,First,Second
, "Game time",100:17,48:02,52:14
, "Effective Playing Time",52:45,29:53,22:51
, "Home EPT",30:31,17:57,12:34
, "Away EPT",22:13,11:56,10:17

ID,Player,Minutes,Distance,Walking,Jogging,Running,"High Speed Running",Sprinting,"No. of
High Intensity Runs","Top Speed","Average Speed","Distance TIP","HSR Distance TIP","Sprint
Distance TIP","No. of High Intensity Runs TIP","Distance OTIP","HSR Distance OTIP","Sprint
Distance OTIP","No. of High Intensity Runs OTIP","Distance BOP","HSR Distance BOP","Sprint
Distance BOP","No. of High Intensity Runs BOP"
103955,"Raheem
Sterling",87:14,8445.31,3325.53,3218.2,1115.58,493.73,292.27,78,33.08,5.81,3726.59,292.25,23
9.71,51,2474.0,182.82,52.57,26,2244.72,18.66,0,1
116594,"N'Golo
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Kante",55:50,5553.81,4420.67,4524.53,5976.27,655.86,595.78,53,59.59,5.68,5529.57,557.15,122.1,17,3118.59,117.11,74.48,34,2135.15,11.61,1,1,
```